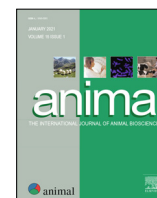




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Effects of rearing systems on production performance, antioxidant capacity and immune status of meat ducks at different ages



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ABSTRACT

As potential substitutes for traditional free-range rearing system, floor-rearing system (**FRS**) and net-rearing system (**NRS**) are the current predominant dryland duck rearing systems. However, the influence of these two systems on production performance and duck health is poorly understood. In this study, a 2×3 factorial arrangement with two rearing systems (FRS and NRS) and three ages (4w, 8w and 13w) was conducted to study the effects of FRS and NRS on production, antioxidant capacity and immune status of Nonghua ducks. The production performance was mainly affected by the effect of rearing systems at 8w. Body weight, average daily gain, eviscerated weight and semi-eviscerated weight were higher in NRS ducks at 8w, but carcass yield at 8w and 13w was decreased ($P < 0.05$). Lipid deposition was enhanced in NRS and higher sebum and abdominal fat yields were seen at 8w and 13w ($P < 0.05$). NRS resulted in developmental retardation of the liver at 4w and decreased gizzard index at all ages ($P < 0.05$). Antioxidant capacity indicators were unaffected by rearing systems ($P > 0.05$), however, in NRS, slightly better antioxidant capacity was seen at 4w, while glutathione peroxidase (**GSH-Px**) activity was higher at 13w ($P < 0.05$). NRS ducks had higher thymus weight at 8w and higher spleen weight at 13w ($P < 0.05$). Immune cytokines were extensively affected by rearing system ($P < 0.05$) and higher levels of interferon- γ , interleukin-1 β , interleukin-4 and immunoglobulins were seen in NRS ducks. Serum biochemical parameters (aspartate aminotransferase (**AST**), alanine aminotransferase (**ALT**) and alkaline phosphatase (**ALP**)) showed that NRS was better for liver health, and in the liver of FRS ducks, **ALP** was higher at 13w, and both **ALP** and interferon- γ were higher at 13w than at 4w and 8w ($P < 0.05$). In conclusion, this study showed that NRS was, to an extent, conducive to production performance and duck liver health, but compared to FRS, defects were seen in visceral organ development and lipid deposition. Although antioxidant capacity was not significantly affected, NRS ducks may have better antioxidant capacity at the early breeding stage, and GSH-Px activity was increased for scavenging excess free radicals at the later one. NRS increased serum levels of interferon- γ , interleukin-1 β , interleukin-4 and immunoglobulins and promoted thymus and spleen development, thus improving duck immune function. These findings will provide a reliable reference for selecting a rearing system.

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Implications

The rearing system is a significant nongenetic factor that affects production benefits and poultry health in duck husbandry, and understanding the effects of two major dryland rearing systems on production performance and health status of meat ducks is of great significance to the industry. This study found that growth

performance, carcass traits, immune functions and antioxidant capacity of Nonghua ducks are promoted, to an extent, in net-rearing system but visceral organ development and lipid deposition were better in floor-rearing system. These findings will provide valuable information for improving production benefits and poultry health during rearing.

Introduction

China is the world's major producer and consumer of broiler ducks (Bai et al., 2020) and rearing practices have evolved from

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free-range to intensive to improve production efficiency (Zhang et al., 2018c), which has led to duck husbandry moving from water associated to terrestrial (Qian et al., 2017). Traditional rearing systems have extensive water demands (Li et al., 2017a; Bai et al., 2020), and the lower water needs of dryland systems can alleviate environmental issues, such as river and lake usage, and may also reduce incidence of intestinal disease outbreaks caused by water-borne pathogens (Zhao et al., 2019a).

Dryland rearing systems are generally classified into cage-rearing system (CRS), floor-rearing system (FRS), and net-rearing system (NRS). Although CRS makes full use of breeding house space, the ducks are vulnerable as individual activities are constrained (Bai et al., 2020). FRS involves laying a 5–10 cm thick pad of straw, sawdust or wood shavings onto a cement floor (Wang et al., 2015; de Almeida et al., 2017) to absorb excess moisture, but there are associated difficulties in rearing management. Plastic and wire floor NRSs are used worldwide, and stainless wires, although they have sharp edges, are more durable than plastic floors. Both create a more hygienic environment than other systems because the ducks have reduced contact with their excreta (Zhang et al., 2018a and 2018c).

The rearing systems may be either positively or negatively correlated with environmental impact and physiological characteristics of ducks during rearing (Fraley et al., 2013). Previous studies on rearing systems have indicated that they can have a range of effects on the ducks, and that production performance is a key concern. A study on Pekin ducks showed that dryland-reared ducks had higher BW than those in an intensive rearing system with a swimming pool (Erişir et al., 2012). When comparing NRS and FRS, Zhang et al. (2018a) found that NRS ducks had higher final BW, average daily gain (ADG), average daily feed intake (ADFI) and lower feed conversion rate (FCR), and that significant changes in serum biochemical parameters could be seen. Fraley et al. (2013) compared Pekin ducks in plastic floor net-rearing with litter floor-rearing, and showed that NRS ducks had higher ADG, but saw no discernible effects on duck body condition. A study on Cherry Valley ducks reported higher ADG and lower FCR in FRS, and concluded that it was a better duck rearing system than NRS (Chen et al., 2015). The differences between these findings could be due to the different breeds, duck physiological status and sampling weeks, used in the studies. Therefore, a systematic evaluation of rearing system effects on duck production performance, including growth performance, carcass traits, visceral organ development over time and physiological status, is essential.

Since animal health status is generally correlated with production performance, the selection of an appropriate rearing system in modern intensive poultry production is crucial (Rehman et al., 2017). Health status is associated with individual immunity and antioxidant capacity and studies have shown that the rearing system can significantly influence the immune response to avian influenza vaccine in different breeds (El-Edel et al., 2015), and affect the expression of key immune genes such as toll-like receptor 7 (TLR7) (Kolluri et al., 2014). One study on Shaoxing ducks reported that NRS had positive impacts on duck health (Zhao et al., 2019a). It has been observed that waterfowl in FRS suffered from temperature fluctuations (Chen et al., 2015; Zhang et al., 2018a) and that either heat or cold stress caused by these temperature changes could lead to oxidative stress, changes in antioxidant defence system, and may threaten the health of the animals. The potential for FRS and NRS to affect duck health may be reflected in the changes in immune status and antioxidant capacity, however, few studies have investigated this.

The aim of this study was to examine the effects of FRS and NRS on production performance, antioxidant capacity and immune status of Nonghua ducks. This would reveal the physiological impacts

of predominant dryland rearing systems on meat ducks and provide a reliable reference for rearing system selection.

Material and methods

Animals and experiment design

This study utilized a 2×3 factorial arrangement with two different rearing systems (FRS and NRS) and ducks at three different slaughtering ages (4w, appropriate breeding age for meat-type ducks; 8w, market age; 13w, market age). The Nonghua duck is a new locally developed Sheldrake crossbreed with improved roughage tolerance, strong disease resistance, excellent growth performance and superior carcass traits. A total of 1 000 one-day-old Nonghua ducks obtained from one commercial hatchery (HJ12096, Beijing Haijiang incubation equipment manufacturing Co., Ltd., Beijing, China) were wirefloor brooded with a commercial starter diet (Supplementary Table S1) for two weeks. Seven hundred and twenty 14-day-old ducks with a male/female ratio of 1:1 were selected by similar initial BW, then randomly allocated into FRS and NRS (10 replicates of 18 ducks each) with males and females separately (namely, 180 ducks in each group). FRS ducks were reared on a concrete floor with 5 cm sawdust bedding, and bedding materials were added appropriately during the experiment period and cleaned weekly. NRS ducks were reared on a stainless wire mesh bed with 1.0 cm diameter mesh holes at a height of 50 cm above the ground, with excreta cleaned weekly. All duck groups were reared in a well-ventilated house with the same environmental conditions and natural light at the experimental waterfowl breeding farm in Sichuan Agricultural University (Sichuan, China). Temperature was set at 31 °C during first week then gradually decreased to 25 °C at 14 d of age, after which it was kept at approximately 15–20 °C. Stocking density was kept at 5 ducks/m². Experimental diets (Supplementary Table S1) were formulated in accordance with NY/T 2122-2012. The same diet and water were provided *ad libitum* to ducks during the experimental period up to 8w, and from 8 to 13w, ducks were only fed during daytime. Feed intakes were recorded weekly and the ADFI calculated (Supplementary Table S2).

Sample collection and production performance measurement

All ducks were weighed at one-day-old and at the end of 4th, 8th and 13th week, three ducks from each replicate (a total of 30 ducks per group) were randomly selected, fasted for 12 h, and then live BW and ADG were measured. Blood samples from all ducks were collected by venepuncture with 0.7 * 25 mm disposable venous blood sampling needles (Jiangxi FUERKANG Industrial Group Co., Ltd., Jiangxi, China) at 9:00 am. Around 5 mL of blood was collected by using 5-mL vacuum anticoagulant-free tubes (Jiangsu Kangjian Medical Apparatus Co., Ltd., Jiangsu, China). Blood was left to stand for 24 h to allow clotting and stratification. The supernatant phase (the light-yellow fluid) was collected and stored at –20 °C for later use in the determination of duck antioxidant capacity and immune status.

After slaughter, 10 carcass traits were measured according to NY/T 823-2004, including carcass weight, semi-eviscerated weight, eviscerated weight, sebum fat weight, abdominal fat weight, carcass yield, semi-eviscerated yield, eviscerated yield, sebum fat yield and abdominal fat yield. The visceral and immune organs (thymus, spleen and bursa of Fabricius) were separated and collected to evaluate duck immune status. Visceral organ parameters including, heart weight, heart index, liver weight, liver index, gizzard weight, gizzard index, proventriculus weight and proventricu-

lus index, were measured to assess production performance. And the organ index was calculated as follows: organ index = the organ weight (g)/individual BW (g). After production performance was determined, 20.0 g of liver samples from each male ducks were collected for gene expression analysis.

Determination of serum biochemical parameters and antioxidant capacity indicators

Seven of the 30 serum samples from each group were randomly selected for serum parameter analysis. The following biochemical parameters were measured by automatic biochemical analyser (Chemray 240, Shenzhen Rayto Life and Analytical Sciences Co., Ltd., Shenzhen, China); alanine aminotransferase (**ALT**), aspartate aminotransferase (**AST**), alkaline phosphatase (**ALP**), total proteins, albumin and globulin. Serum antioxidant capacity indicators were measured using commercial assay kits (Nanjing Jincheng Bioengineering Institute, Nanjing, China) with the thiobarbituric acid colorimetry (TBA) method for malondialdehyde hydroxylamine method for total-superoxide dismutase activity, and the colorimetric method for glutathione peroxidase (**GSH-Px**) ([Supplementary Materials S1–S3](#)).

Determination of immune status

Immune status was evaluated by immune organ development and serum immune cytokine profile. The following six immune organ parameters were measured: spleen weight, spleen index, thymus weight, thymus index, bursa of Fabricius weight and bursa of Fabricius index. The following serum immune cytokines were measured using commercial ELISA kits (Wuhan Chundu Biological Technology Co., Ltd., Wuhan, China) in accordance with manufacturer's instructions ([Supplementary Materials S4](#)); immune globulin-A, immune globulin-G, immune globulin-M, interleukin-1 β , interleukin-4, interleukin-6 and interferon- γ .

Total RNA extraction and cDNA synthesis

Four liver samples from male ducks in different rearing systems and different ages were used to determine expression of genes related to immune function and antioxidant capacity. Total RNA was isolated from the frozen liver tissues using Trizol reagent (TaKaRa Bio Inc, Dalian, China) following the manufacturer's instructions and extracted RNA concentration and integrity was measured by electrophoresis and NanoDrop 2000. Based on the

clear bands for 5S rRNA, 18S rRNA and 28S rRNA visible under ultraviolet light and the A260:A280 ratio range within 1.8–2.1, meant that the extracted RNA was deemed acceptable for reverse transcription. Reverse transcription was performed using Primer-Script™ RT reagent Kit with gDNA Eraser (TaKaRa Bio Inc, Dalian, China) according to the manufacturer instructions. cDNA products were stored at -20°C prior to quantitative real-time PCR analysis.

Quantitative real-time PCR analysis

Primers were designed based on GenBank sequences using Primer Premier 5.0, with β -actin and GAPDH taken from [Li et al. \(2017b\)](#) and being used as reference genes. Primers were obtained from Beijing Tsingke Biotechnology Co., Ltd. (Sichuan, China) and are shown in [Table 1](#).

Quantitative real-time PCR was performed on a CFX96™ Real-Time PCR system (Bio-Rad, CA, U.S.A.) using SYBR Premix ExTaq™ II (TaKaRa Bio Inc, Dalian, China). A total PCR reaction volume of 12.5 μL consisted of 6.25 μL of SYBR Premix ExTaq™ II, 4.25 μL ddH₂O, 1.0 μL cDNA template and 0.5 μL of each primer. The PCR conditions used were as follows: 3 min predenaturation at 95°C , followed by 40 cycles of 10 sec denaturation at 95°C , 30 s annealing at primers optimum reaction temperature and 30 s extension at 72°C . A melting curve was performed to verify that the amplified PCR product was single. Every sample was tested three times and adopted with standard deviations of threshold cycle (CT) below 0.5. The relative expression level of each target gene was calculated using the $2^{-\Delta\Delta\text{CT}}$ method against reference genes.

Statistical analysis

Experimental data were analysed using the GLM procedure in SPSS 22.0. Each group was an experimental unit. Analysis of variance was performed using a 2×3 factorial design to test the effects of rearing system and age on production performance, antioxidant capacity, immune status and liver gene expression. All parameters were analysed using the following univariate linear model:

$$Y_{ijk} = \mu + \text{rearing system}_i + A_j + \text{rearing system} \times A_{ij} + e_{ijk}$$

where Y_{ijk} is the observed trait for duck k in rearing system i at age j , μ is the overall mean, rearing system_i is the fixed effect of rearing system ($i = \text{FRS}$ and NRS), A_j is the fixed effect of sampling age ($j = 4\text{w}$, 8w , and 13w), $\text{rearing system} \times A_{ij}$ is the effect of the interactions between rearing system and sampling age, and e_{ijk} is the

Table 1
Primer pairs used for real-time PCR of Nonghua ducks in this study.

Gene ¹	Primer sequences (5'–3')	GenBank accession	Amplicon, bp	Annealing temperature, $^{\circ}\text{C}$
β -actin	Forward: GCTATGTCGCCCTGGATTTC Reverse: CACAGGACTCCATACCAAGAA	NM_001310421	168	60
GAPDH	Forward: AAGGCTGAGAATGGGAAAC Reverse: TTCAGGGAATGTCATACTTC	XM_027449740	254	54
SOD1	Forward: TGTCTGATGGAGATCATGGCTTCC Reverse: CCTTGCCAGGTCGTCGCTTTT	KU048808	271	68
ALB	Forward: CTACAATGATTTGAAGGAGGAG Reverse: CTGCGAATGAAGCCTGGGATA	XM_005012746	664	62.5
IFN- γ	Forward: TCATACTGAGCCAGATTGTAC Reverse: CTGTAGTCATTGGAAGGGTATT	AF087134	129	56
GOT1	Forward: CTAAGGATGGCCGCAAGAGG Reverse: CTCAGGCTCGGAGACAAAG	XM_027460734	273	67.5
ALP	Forward: GCTCTGCTTGGCATCGCTGGTC Reverse: GGCTTTGACCCGAGAGGTAGGC	XM_027443431	335	70

¹ GAPDH: glyceraldehyde 3-phosphate dehydrogenase; SOD1: superoxide dismutase 1; ALB: albumin; IFN- γ : interferon-gamma precursor; GOT1: glutamic-oxaloacetic transaminase 1; ALP: alkaline phosphatase.

residual error. Duncan's multiple-range test was conducted between the six groups (4w-FRS, 4w-NRS, 8w-FRS, 8w-NRS, 13w-FRS and 13w-NRS) to test the significance of the effects used in the statistic model and $P < 0.05$ was considered statistically significant. Values were presented as mean with SEM.

Results

Production performance

Growth performance and carcass trait results are shown in Table 2. BW and ADG were affected by rearing system and the interactions between rearing system and age ($P < 0.05$). BW increased with age and both BW and ADG were higher in NRS than FRS ducks at 8w ($P < 0.05$). With regard to carcass traits, carcass yield, eviscerated yield and semi-eviscerated yield of Nonghua ducks exhibited a consistent tendency higher than 85%, 70% and nearly 80% at different ages in both FRS and NRS. carcass weight, sebum fat weight, sebum fat yield, abdominal fat weight and abdominal fat yield were affected by rearing system ($P < 0.05$). carcass yield, eviscerated weight, eviscerated yield, semi-eviscerated weight, sebum fat weight, sebum fat yield, abdominal fat weight and abdominal fat yield were affected by the interactions between rearing system and age ($P < 0.05$). carcass yield was lower in NRS ducks at 4w and 8w ($P < 0.05$), while carcass weight, eviscerated weight and semi-

eviscerated weight were higher in NRS ducks at 8w ($P < 0.05$). sebum fat weight, sebum fat yield, abdominal fat weight and abdominal fat yield were significantly higher in NRS ducks at 8w ($P < 0.05$), and sebum fat yield, abdominal fat weight and abdominal fat yield were also higher at 13w ($P < 0.05$). Visceral organ results are shown in Table 3. Liver index, gizzard weight and gizzard index were all directly affected by rearing system ($P < 0.05$), while heart weight, liver weight, liver index, gizzard weight and gizzard index were affected by the interactions between rearing system and age ($P < 0.05$). Liver weight and liver index were significantly higher in FRS ducks at 4w ($P < 0.05$). Gizzard weight was higher in FRS ducks at 4w and 13w ($P < 0.05$), and gizzard index was higher in FRS ducks at all ages ($P < 0.05$).

Serum biochemical parameters and antioxidant capacity indicators

Serum biochemical parameter results are shown in Fig. 1(a). ALT, ALP, total protein, albumin and globulin were affected by rearing system ($P < 0.05$), while total protein, albumin and globulin were also affected by the interactions between rearing system and age ($P < 0.05$). With regard to liver health, ALP was significantly lower in NRS ducks at 4w and 8w ($P < 0.05$), while ALT was lower in NRS ducks at 8w and 13w ($P < 0.05$). total protein and globulin were significantly higher in NRS ducks but albumin was lower than FRS ducks at 13w ($P < 0.05$). Although neither the

Table 2

Effects of FRS and NRS on growth performance and carcass traits of Nonghua ducks.

Items ¹	Week 4		Week 8		Week 13		SEM	P-value ²		
	FRS	NRS	FRS	NRS	FRS	NRS		RS	Age	RS × Age
BW (g)	1 195.3 ^c	1 186.3 ^c	2 448.3 ^b	2 663.7 ^a	2 679.5 ^a	2 650.1 ^a	39.156	0.021	<0.001	<0.0001
ADG (g)	41.1 ^b	40.8 ^b	42.9 ^b	46.8 ^a	29.0 ^d	28.7 ^d	0.443	0.019	<0.001	<0.001
CW (g)	1 077.9 ^d	1 026.6 ^d	2 153.1 ^c	2 318.1 ^b	2 407.3 ^a	2 354.2 ^{ab}	33.729	0.367	<0.001	<0.001
CY (%)	89.4 ^a	86.9 ^{bc}	87.6 ^b	86.5 ^c	89.2 ^a	88.7 ^a	0.142	<0.001	<0.001	0.005
EW (g)	867.0 ^d	839.3 ^d	1 845.3 ^c	2 013.2 ^b	2 087.5 ^a	2 058.2 ^{ab}	30.583	0.064	<0.001	<0.001
EY (%)	71.6 ^c	71.0 ^c	75.0 ^b	75.5 ^b	77.9 ^a	77.6 ^a	0.191	0.566	<0.001	0.207
SEW (g)	972.1 ^d	933.4 ^d	1 995.1 ^c	2 185.2 ^b	2 269.8 ^a	2 230.2 ^{ab}	32.604	0.088	<0.001	<0.001
SEY (%)	80.2 ^c	78.9 ^d	81.3 ^b	81.3 ^b	84.3 ^a	84.1 ^a	0.162	0.060	<0.001	0.061
SPW (g)	201.5 ^d	199.0 ^d	410.2 ^c	513.0 ^a	481.5 ^b	509.3 ^{ab}	8.317	<0.001	<0.001	<0.001
SFY (%)	23.2 ^{cd}	23.7 ^{bc}	22.1 ^d	25.5 ^a	23.1 ^{cd}	24.7 ^{ab}	0.184	<0.001	0.520	0.004
AFW (g)	7.5 ^c	8.1 ^c	25.1 ^d	44.3 ^b	38.1 ^c	50.5 ^a	1.106	<0.001	<0.001	<0.001
AFY (%)	0.9 ^d	0.9 ^d	1.2 ^c	2.0 ^b	1.8 ^b	2.4 ^a	0.042	<0.001	<0.001	<0.001

^{a-e}Different lowercase letters indicated that the results of post hoc analysis were significant ($P < 0.05$).

¹ ADG, average daily gain; CW, carcass weight; CY, carcass yield; EW, eviscerated weight; EY, eviscerated yield; SEW, semi-eviscerated weight; SEY, semi-eviscerated yield; SPW, sebum fat weight; SFY, sebum fat yield; AFW, abdominal fat weight; AFY, abdominal fat yield; FRS, floor-rearing system; NRS, net-rearing system; RS, rearing system.

² When P -values were below 0.001, the specific values were omitted and uniformly expressed as <0.001.

Table 3

Effects on visceral organ development of Nonghua ducks at different ages.

Items ¹	Week 4		Week 8		Week 13		SEM	P-value ²		
	FRS	NRS	FRS	NRS	FRS	NRS		RS	Age	RS × Age
Heart weight	8.188 ^c	7.880 ^c	15.873 ^b	16.735 ^a	17.223 ^a	16.769 ^a	0.236	0.864	<0.001	0.010
Heart index	0.682 ^a	0.667 ^{ab}	0.651 ^{abc}	0.636 ^{bc}	0.651 ^{abc}	0.631 ^c	0.005	0.066	0.003	0.972
Liver weight	40.203 ^c	34.620 ^d	55.621 ^a	57.947 ^a	49.655 ^b	51.180 ^b	0.627	0.523	<0.001	<0.001
Liver index	3.345 ^a	2.942 ^b	2.274 ^c	2.213 ^c	1.878 ^d	1.972 ^d	0.037	0.006	<0.001	<0.001
Gizzard weight	43.897 ^b	39.988 ^c	60.876 ^a	59.693 ^a	58.322 ^a	46.690 ^b	0.611	<0.001	<0.001	<0.001
Gizzard index	3.661 ^a	3.389 ^b	2.484 ^c	2.276 ^d	2.240 ^d	1.751 ^e	0.042	<0.001	<0.001	0.020
PW	5.435 ^c	5.395 ^c	7.563 ^{ab}	7.708 ^{ab}	8.098 ^a	7.426 ^b	0.107	0.295	<0.001	0.154
PI	0.453 ^a	0.456 ^a	0.309 ^b	0.293 ^b	0.302 ^b	0.283 ^b	0.006	0.186	<0.001	0.440

^{a-e}Different lowercase letters indicated that the results of post hoc analysis were significant ($P < 0.05$).

¹ The unit of immune organs weight is "g", while the unit of immune organs index is "%". Abbreviations: PW, proventriculus weight; PI, proventriculus index; FRS, floor-rearing system; NRS, net-rearing system; RS, rearing system.

² When P -values were below 0.001, the specific values were omitted and uniformly expressed as <0.001.

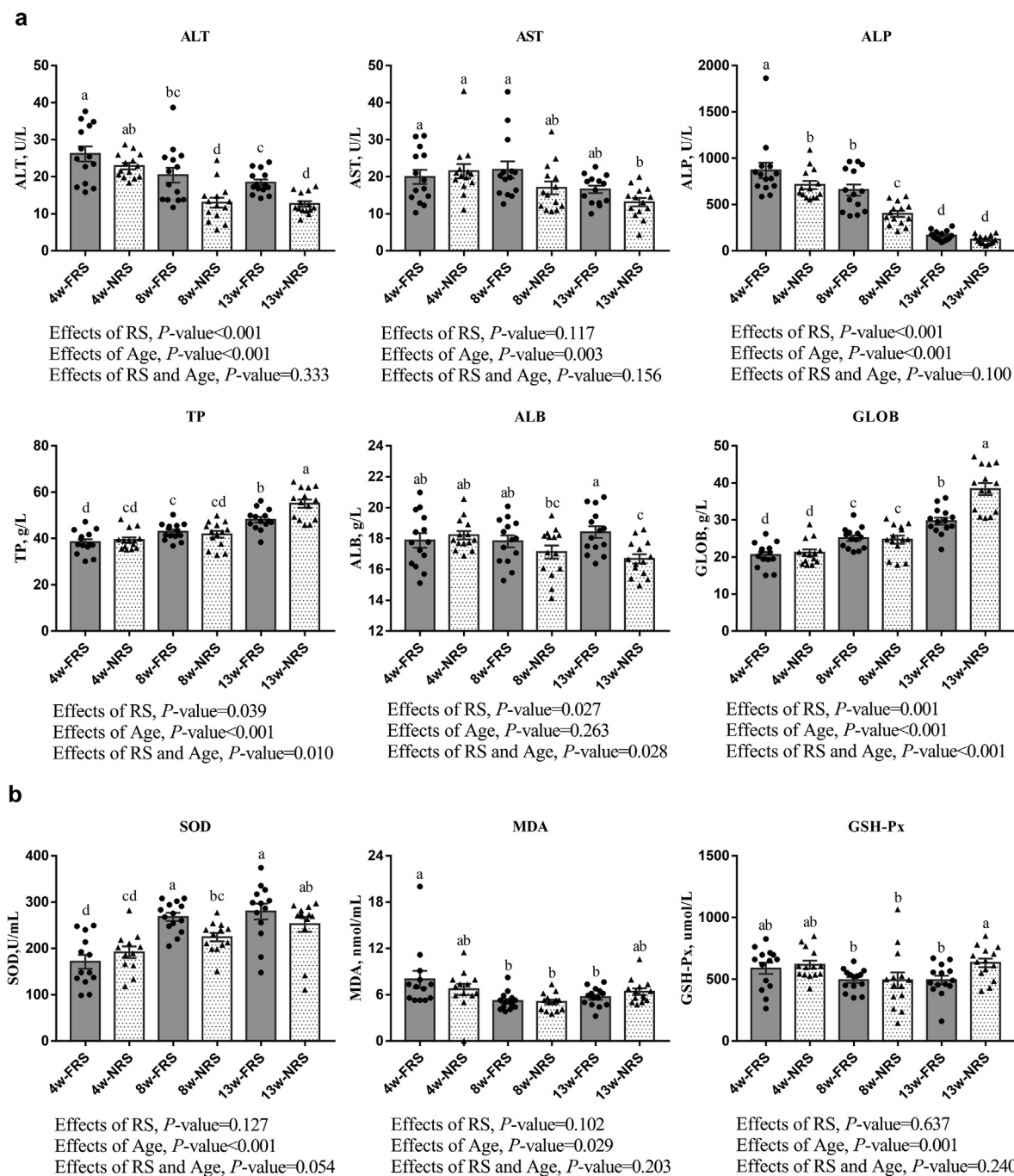


Fig. 1. Effects of FRS and NRS on (a) serum biochemical parameters and (b) antioxidant capacity indicators of Nonghua ducks at different ages. Data displayed as mean $\pm$ SEM. For both FRS and NRS, $n = 14$. The P -values of the main effects of RS, age and their interactions are below the figures, and when a P -value was below 0.001, the specific values were omitted and were uniformly expressed as <0.001. Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; ALP, alkaline phosphatase; TP, total proteins; ALB, albumin; GLOB, globulin; MDA, malondialdehyde; SOD, superoxide dismutase; GSH-Px, glutathione peroxidase; FRS, floor-rearing system; NRS, net-rearing system; RS, rearing system. ^{a-d}Lowercase letters indicate that post hoc analysis results are significant ($P < 0.05$).

rearing system nor the interactions had a significant impact on antioxidant capacity indicators, a lower malondialdehyde content and higher GSH-Px and superoxide dismutase activity were found in NRS ducks at 4w ($P > 0.05$) (Fig. 1b). In NRS ducks, superoxide dismutase activity was decreased at 8w ($P < 0.05$), and GSH-Px activity increased at 13w ($P < 0.05$).

Immune organ development and serum immune cytokine profiles

Immune organ and cytokine results are shown in Table 4 and Fig. 2, respectively. Regarding immune organs, spleen weight,

bursa weight and bursa index were significantly affected by rearing system ($P < 0.05$), while spleen weight, spleen index and thymus weight were significantly affected by the interactions ($P < 0.05$). In NRS ducks, Bursa index was significantly lower at 4w and 8w ($P < 0.05$), thymus weight was significantly higher at 8w ($P < 0.05$), and spleen weight and index were significantly higher at 13w ($P < 0.05$). All immune cytokine indicators were significantly affected by rearing system ($P < 0.05$), and immune globulin-G was also affected by the interactions ($P < 0.05$). In NRS ducks, interleukin-1 β was consistently higher at all ages ($P < 0.05$), interleukin-4 was significantly higher but interleukin-6

Table 4
Effects of FRS and NRS on immune organ development of Nonghua ducks.

Items ¹	Week 4		Week 8		Week 13		SEM	P-value ²		
	FRS	NRS	FRS	NRS	FRS	NRS		RS	Age	RS × Age
Spleen weight	1.287 ^c	1.188 ^c	1.895 ^b	1.840 ^b	1.898 ^b	2.576 ^a	0.049	0.050	<0.001	<0.001
Spleen index	1.079 ^a	0.986 ^a	0.772 ^b	0.684 ^b	0.726 ^b	1.001 ^a	0.021	0.450	<0.001	<0.001
Thymus weight	4.019 ^c	4.148 ^c	8.502 ^b	10.358 ^a	2.963 ^d	2.500 ^d	0.214	0.088	<0.001	0.002
Thymus index	3.338 ^b	3.397 ^{ab}	3.475 ^{ab}	3.876 ^a	1.107 ^c	0.962 ^c	0.093	0.503	<0.001	0.238
Bursa weight	2.478 ^b	2.197 ^b	3.193 ^a	2.955 ^a	1.367 ^c	1.128 ^c	0.059	0.005	<0.001	0.954
Bursa index	2.076 ^a	1.824 ^b	1.309 ^c	1.099 ^d	0.499 ^e	0.405 ^e	0.043	<0.001	<0.001	0.515

^{a-e}Different lowercase letters indicated that the results of post hoc analysis were significant ($P < 0.05$).

¹ The unit of immune organs weight is “g”, while the unit of immune organs index is “%”. Abbreviations: FRS, floor-rearing system; NRS, net-rearing system; RS, rearing system.

² When P -values were below 0.001, the specific values were omitted and uniformly expressed as <0.001.

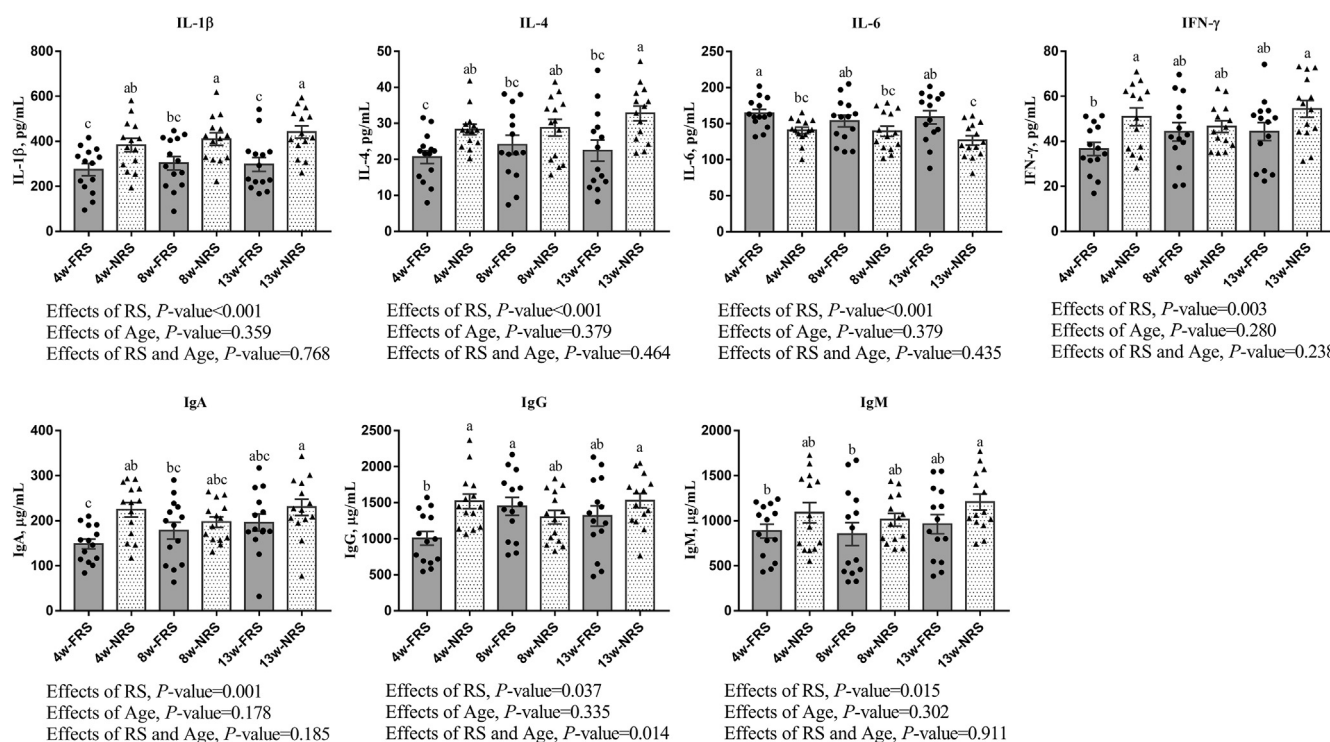


Fig. 2. Effects of FRS and NRS on serum immune cytokines of Nonghua ducks at different ages. Data displayed means $\pm$ SEM. For both FRS and NRS, $n = 14$. The P -values of main effects of RS, age and their interactions are below the figures and when a P -value is below 0.001, the specific values were omitted and uniformly expressed as <0.001. Abbreviations: IL, interleukin; IFN- γ , interferon- γ ; Ig, immune globulin; FRS, floor-rearing system; NRS, net-rearing system; RS, rearing system. ^{a-c}Lowercase letters indicate that post hoc analysis results are significant ($P < 0.05$).

was lower at 4w and 13w, and interferon- γ , immune globulin-A and immune globulin-G were only higher at 4w ($P < 0.05$).

Liver gene expression

Gene expression analyses are shown in Fig. 3. *Albumin* was directly affected by rearing system ($P < 0.05$) and was consistently lower in NRS ducks at all ages. Other genes were neither affected by rearing system nor affected by the interactions ($P > 0.05$). In 4w-FRS ducks, *albumin* was higher than both 8w and 13w ($P < 0.05$) and was significantly higher than NRS ducks at 4w ($P < 0.05$). In 13w-FRS ducks, *ALP* was significantly higher than 4w and 8w ($P < 0.05$) and was also significantly higher than NRS ducks at 13w ($P < 0.05$). *interferon- γ* was higher in 13w-FRS than 4w-FRS and 8w-FRS ($P < 0.05$).

Discussion

Based on the higher final BW and ADG, and lower FCR of waterfowls in NRS, studies on Yangzhou geese and Chaohu ducks have reported that waterfowls reared in NRS exhibited better growth performance than in FRS (Liu et al., 2011; Zhang et al., 2018a). In Moulard ducks, although FCR was not significantly affected, the higher final BW and ADG in NRS compared to sawdust-FRS and sand-FRS also showed that NRS resulted in better growth performance (Mohammed et al., 2019). However, in Cherry Valley ducks reared in NRS, Chen et al. (2015) observed a lower ADG and higher FCR, thus reaching an opposite conclusion that FRS resulted in better growth performance. These differing conclusions could be attributed to the different breeds and sampling ages used in these studies. To clarify the effects of FRS and NRS on meat ducks in different breeding stages, in this study, three key ages were used to

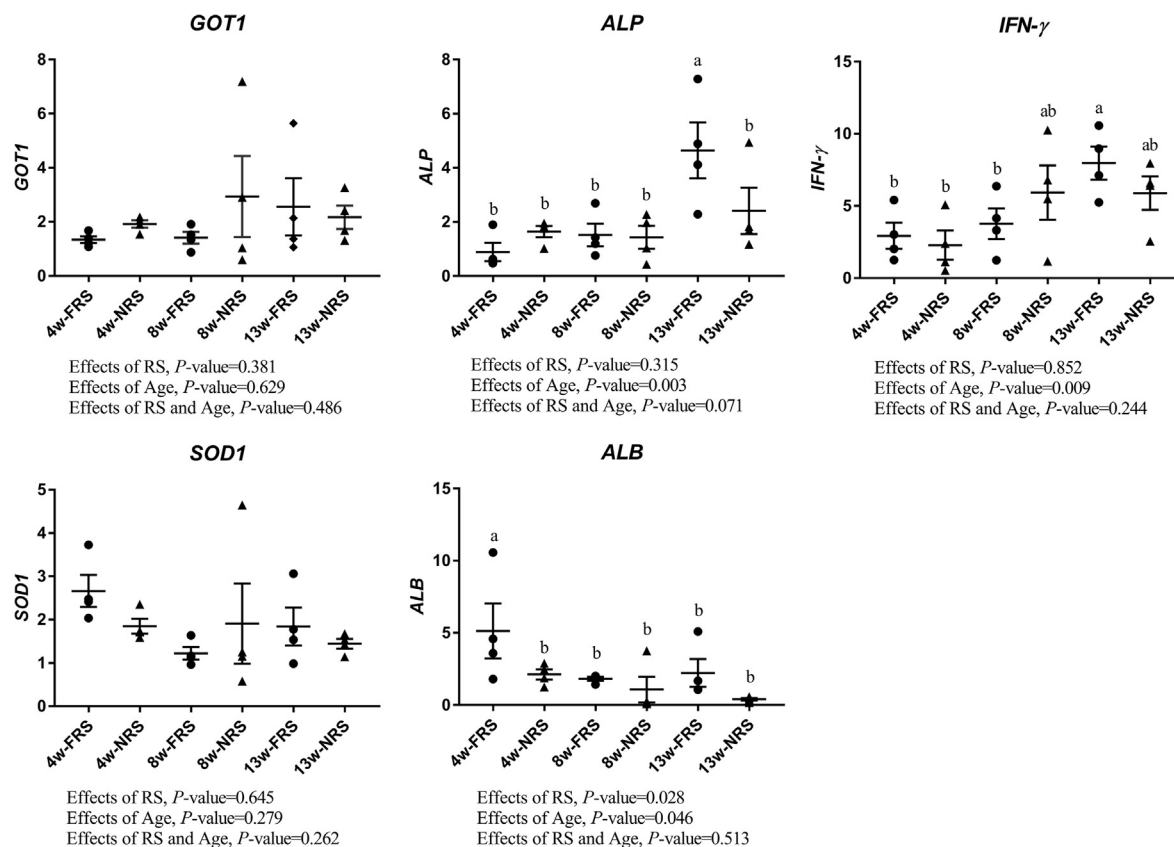


Fig. 3. Effects of FRS and NRS on liver gene expressions of Nonghua ducks at different ages. Data were displayed as mean $\pm$ SEM. For both FRS and NRS, $n = 4$. The P -values of the main effects of RS, age and their interactions are below the figures. Abbreviations: GOT1, glutamic-oxaloacetic transaminase 1; ALP, alkaline phosphatase; IFN- γ , interferon- γ ; SOD1, superoxide dismutase 1; ALB, albumin; FRS, floor-rearing system; NRS, net-rearing system; RS, rearing system. ^{a,b}Lowercase letters indicate that post hoc analysis results are significant ($P < 0.05$).

study the interactions between rearing system and age. Results showed that at 8w, BW and ADG were higher in NRS ducks and that no significant differences were seen at 4w and 13w, indicating that the growth performance of NRS was better before market age.

With regard to carcass traits, the average carcass yield, eviscerated yield and semi-eviscerated yield at 8w and 13w under either FRS or NRS were higher than 85%, 75% and 80%, indicated that the Nonghua ducks had outstanding carcass traits compared to Pekin, Cherry Valley, White Muscovy and Jingjiang ducks (a native Sheldrake breed in China) at similar ages (Wu et al., 2010; Zeng et al., 2015; Ling et al., 2018; Zhang et al., 2018b). A previous study on Chaohu ducks saw that NRS resulted in a lower eviscerated yield but higher abdominal fat yield (Zhang et al., 2018a). This is similar to the results of this study which showed that in NRS ducks, carcass yield was lower than FRS ducks at 4w and 8w, while carcass weight, eviscerated weight, semi-eviscerated weight, sebum fat weight, abdominal fat weight, sebum fat yield and abdominal fat yield were all higher at 8w, and sebum fat yield and abdominal fat yield were also higher at 13w, indicating that NRS was conducive to carcass traits to an extent. As age increased, most significant differences in carcass traits faded, but lipid deposition was consistently enhanced in NRS. This agrees with Liu et al. (2011) who reported that Yangzhou geese exhibited higher subcutaneous fat thickness and higher abdominal fat yield in NRS compared to FRS. Visceral organ development is closely associated with duck growth, development and health (Jiang et al., 2018). The results in this study showed that GW and GI were generally lower in NRS across different ages, indicating that FRS ducks had better gizzard development, which agrees with Wang et al. (2015), who also found that FRS birds had better gizzard development than both CRS

and NRS. It has been proposed that this is because NRS ducks cannot consume any sand from the floor, and that adding large particles and structural components to diets could stimulate gizzard function and development (Svihus et al., 2013; Xu et al., 2017). In this study, the liver weight and liver index of NRS ducks were lower at 4w, indicative of a retardation of liver development at the early breeding stage in NRS. Therefore, while NRS improved the carcass weight before the 8 wk, there were also defects in carcass yield. FRS ducks exhibited better gizzard and liver development at the early breeding stage, while NRS enhanced lipid deposition at the later one.

As the liver is the major site of metabolism and detoxification, this developmental retardation seen in NRS ducks may influence serum biochemical parameters. AST, ALT and ALP are biomarkers used to evaluate liver health, because hepatocytes will secrete these enzymes into the blood stream when the liver is damaged (Hang et al., 2019). Our results showed these parameters were mostly lower in NRS ducks at different ages, suggesting that NRS was better for liver health despite the enhanced lipid deposition and retardation of liver development. This shows another advantage of NRS over CRS, as ducks that had suffered liver injury once were put into cages (Zhang et al., 2019). Antioxidant capacity prevents cell damage from free radicals and other reactive oxygen species and helps to avoid oxidative stress (the imbalance between reactive oxygen species and antioxidants) (Ruan et al., 2019). malondialdehyde levels and superoxide dismutase and GSH-Px activities are effective indicators of the duck's antioxidant status (Zhao et al., 2019b). A study on Shaoxing ducks showed that rearing systems did not have significant impacts on malondialdehyde, superoxide dismutase, catalase, total antioxidant capacity and

GSH-Px activities in the liver (Zhang et al., 2019). Another study which only determined the serum total antioxidant capacity showed no significant difference between FRS and NRS (Zhao et al., 2019a). Similar to this study, the rearing system did not significantly affect malondialdehyde, superoxide dismutase and GSH-Px, however, malondialdehyde content was lower and superoxide dismutase and GSH-Px activity was higher in NRS at 4w, suggesting that NRS ducks might have a better antioxidant capacity at this early stage. Superoxide dismutase activity decreased in NRS at 8w and was relatively lower than FRS ducks at 13w, while GSH-Px activity was significantly higher and malondialdehyde content was relatively higher than FRS ducks at 13w. These results imply that GSH-Px is responsible for scavenging excess free radicals at the later breeding stage.

Regarding immune function, Xi et al. (2015) reported no significant differences between NRS and FRS in the immune organ indexes of Cherry Valley ducks. However, in this study, NRS ducks had a higher thymus weight at 8w as well as a higher spleen weight and spleen index at 13w, indicating that in NRS ducks, immune organ development was promoted. The decreased bursa index in NRS ducks at 4w and 8w implied a faster degradation of bursa. This study also showed a higher thymus index in NRS ducks at 4w and 8w, although the results were not significant. This was comparable to the study of Zhao et al. (2019a), who reported that NRS ducks had lower mortality rates and higher thymus index than FRS ducks. However, in contrast with their report that the interleukin-1 β and immune globulin-G profiles of Shaoxing ducks were not significantly affected by NRS (Zhao et al., 2019a), immune cytokines were extensively affected by rearing system in our study. In general, the higher interferon- γ , interleukin-1 β , interleukin-4 and immunoglobulins levels in NRS ducks indicated a better immune status. IL-1 (Dinarello, 2009) and interleukin-4 (Yitbarek et al., 2013) contribute to T helper type-2 cell activity, and interferon- γ contributes to T helper type-1 cell activity (Yitbarek et al., 2015). The higher interferon- γ , interleukin-1 β and interleukin-4 levels observed during this study suggest that T helper type-1 and T helper type-2 cell activity may be enhanced in NRS. Because IL-1 can act as a growth factor for thymocytes (Dinarello, 2009), the thymus enlargement seen at 8w may be associated with elevated interleukin-1 β levels. Faster bursa degradation and lower interleukin-6 levels suggest that NRS ducks may have lower immunological requirements. This could be because the direct dropping of duck excreta in NRS means that ducks have less opportunity to contact with it. Wang et al. (2003) and de Almeida et al. (2017) reported that FRS had higher microbial contamination and higher concentrations of NH₃ and CO₂ in the air than NRS, and so the better air quality in NRS may also help further reduce the immune system burden in the ducks. The higher immunoglobulin levels in NRS may be associated with elevated interleukin-1 β , which promotes IL-2 and enhances immunoglobulin secretion. In terms of serum proteins, NRS ducks had higher globulin and total protein but lower albumin levels at 13w, which agrees with a study in Chaohu ducks (Zhang et al., 2018a), which also reported that serum total protein content was improved by NRS compared to FRS. The changes in serum proteins can be described as follows: First, FRS ducks were more active during rearing, enhancing protein synthesis and causing total protein loss in the blood (Rehman et al., 2017). Second, due to the enhanced lipid deposition in NRS, albumin would be used for lipoprotein synthesis thereby reducing its serum levels (van der Vusse et al., 2002). Third, the higher immunoglobulin levels in NRS contributed to the change globulin serum levels, especially the consistent elevation in immune globulin-A, immune globulin-G and immune globulin-M in NRS at 13w. Based on the liver functions of synthesizing serum proteins including albumins (Nikoozad et al., 2014), a consistent trend of lower albumin expression in NRS ducks at

different ages might be responsible for the significantly lower serum albumin profiles at 13w. There was no significant change in *superoxide dismutase 1*, indicating that this antioxidant was not significantly affected by the different rearing systems. Interferon- γ is a pro-inflammatory cytokine, which participates in multiple inflammatory responses (Yitbarek et al., 2015) and so the higher level of interferon- γ expression as well as ALP in 13w-FRS compared to 4w-FRS and 8w-FRS indicates that there is potentially a greater environmental stress in liver at the later breeding stage in FRS. However, the precise regulatory mechanism of this will require further study to elucidate. Therefore, NRS could be seen to improve duck immune function by increasing the levels of certain immune cytokines and promoting the development of immune organs. The liver gene expression profile indicates that ducks might suffer from a potential higher level of environmental stress in FRS over time.

Conclusion

Compared to FRS, NRS was associated with improved growth performance and carcass traits, but was also associated with defects in visceral organ development and lipid deposition. According to the lower serum levels of ALT, AST and ALP as well as liver gene expression, NRS may be better for overall duck liver health compared to FRS. Antioxidant capacity was not significantly affected by rearing system, but indicators showed that NRS ducks may have better antioxidant capacity at the early breeding stage, while at the later one, GSH-Px activity was increased to help scavenge excess free radicals in NRS ducks. NRS increased levels of interferon- γ , interleukin-1 β , interleukin-4 and immunoglobulins in serum and promoted thymus and spleen development to improve duck immune function, which would then promote T helper type-1 and T helper type-2 cell activities.

Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.animal.2021.100199>.

Ethics approval

All protocols used in the current study were authorized by the Institutional Animal Care and Use Committee of Sichuan Agricultural University (Approval No. DKY-B20141401).

Data and model availability statement

None of data were deposited in an official repository.

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Declaration of interest

The authors declare that they have no competing interests.

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